Aashray Jain

Software Developer | Full-Stack & AI/ML

+91-9201876340 | aashrayjain99@gmail.com | github.com/aashrayjain
linkedin.com/in/aashray-jain-b26168119 | aashrayjain.github.io

Summary

Full-Stack & Backend Developer with 4 years of professional experience in Node.js, React, Angular, Python, AWS, REST APIs, and scalable enterprise application development. Experienced in backend architecture, cloud migration, database systems, automation, and performance optimization, with hands-on expertise in AI/ML, Generative AI, RAG, and LLM-based solutions. Proven ability to build production-ready applications and independent technology products, combining software engineering experience with emerging AI technologies. Currently pursuing an M.Tech. in Artificial Intelligence & Machine Learning at NIMS University Rajasthan, Jaipur.

Education

- **M.Tech. in Artificial Intelligence & Machine Learning** — *2026 – 2028*
  NIMS University Rajasthan, Jaipur — Pursuing
- **B.Tech. in Electronics & Communication Engineering** — *2017 – 2021*
  Lakshmi Narain College of Technology, Bhopal, Madhya Pradesh (RGPV Bhopal) — 8.74 CGPA
- **Senior Secondary** — *2017*
  R.S. Memorial H.S. School, Katni, Madhya Pradesh (MP Board) — 81.4%
- **Secondary** — *2015*
  DAV ACC Public School, Katni, Madhya Pradesh (CBSE Board) — 8.0 CGPA

Experience

- **ArihantQ** — *Aug. 2025 – Present*
  Founder (Part-Time and Experimental Venture) — Remote (India)
  - Established a personal AI innovation lab focused on Generative AI, RAG, and LLM applications to build a portfolio of full-stack projects and strategically bridge a career transition.
  - Architected and developed key AI-driven prototypes, including an AI-Powered Portfolio WebApp and a RAG-Powered PDF Question Answering App, showcasing expertise in Python, HuggingFace, and LangChain.
  - Drove self-directed learning and validated technical concepts by actively participating in multiple hackathons (**Lyzr AI Architect Challenge, Elastic Forge the Future, MariaDB Python Hackathon**):
    - Achieved recognition in the Lyzr AI Architect Challenge, **winning a** INR **1000 prize** for the innovative 'AI Cost Optimization Advisor' idea.
    - Attended the **Elastic Search AI company's virtual event in Bengaluru**, staying current with industry trends and networking.
    - An innovation idea in the MariaDB Hackathon received public recognition from the **MariaDB, MariaDB Foundation, CEO, and Community Advocacy**.
  - Conducted due diligence and secured business enablers (**Udyam Registration, Shop and Establishment Act License, and a Current Bank Account**) to provide a legal and professional foundation for an independent venture and continuous skill development.
  - Developed a customer-centric innovation pipeline, translating market needs into a product roadmap with a long-term vision for scaling AI solutions.
  - Architected and developed ArihantQInvoice, a full-stack ERP platform for billing, inventory management, GST invoicing, stock tracking, purchase management, and customer/supplier operations using Node.js, Express.js, SQLite, and Sequelize.
  - Filed software copyright application for ArihantQInvoice Billing and Inventory Management with the Copyright Office, Government of India; application currently under scrutiny.
  - **LinkedIn**: linkedin.com/company/arihantq
- **Tata Consultancy Services** — *Jul 2021 – Jan 2025*
  Systems Engineer — Mumbai & Kochi, India
  - Developed secure, real-time alert analysis systems and Line of Business (LOB) dashboards for a Canadian BFSI client using React, Node.js Lambda, and DynamoDB, improving risk monitoring efficiency by 90% and enhancing operational visibility.
  - Built and maintained scalable REST APIs and backend services using Node.js and Express.js to support secure financial workflows, modular integrations, and compliance-driven enterprise applications.
  - Led a multi-account AWS cloud migration for a large-scale enterprise platform, optimizing infrastructure architecture and reducing operational costs by 70% while improving scalability and resilience.
  - Strengthened enterprise security posture by integrating SAML-based authentication and AWS Cognito for secure user access across applications serving 500K+ users, reducing vulnerabilities and improving compliance by 40%.
  - Automated complex business reporting workflows using Angular, NX Monorepo, Python (Pandas, Flask), and Excel automation, reducing manual effort by 15+ hours per week and improving reporting accuracy.

- Designed interactive analytics dashboards and visualization tools using Angular and D3.js to accelerate business intelligence and decision-making for enterprise stakeholders.
- Collaborated with TCS Rapid Labs on AI/ML and automation PoCs, contributing to innovation initiatives in NLP, Computer Vision, and Generative AI for internal and enterprise use cases.
- Built and evaluated machine learning pipelines for sentiment analysis and classification, strengthening capabilities in data-driven insights and intelligent automation.

- **Tata Consultancy Services** *Mar 2021 – Jun 2021*

Project Intern Remote (India)

- Developed the Asset Procurement & Inventory module for the EquipMax 3.0 project, utilizing Angular, Ionic, and Bootstrap as part of the TCS Accelerator brand project (EXOP-00937).

Projects

- **ArihantQInvoice Billing & Inventory Management System** *Apr 2026*

Node.js, Express.js, SQLite, Sequelize, EJS, Swagger, DOCX Generation

- Designed and developed a production-ready ERP platform for billing, inventory management, purchase tracking, pricing management, stock monitoring, and customer/supplier operations.
- Implemented session-based authentication, invoice lifecycle management, GST and non-GST billing workflows, and automated DOCX invoice generation.
- Built 50+ REST and server-rendered modules covering customer management, supplier management, item master, purchase processing, pricing engine, stock movement tracking, and transport operations.
- Developed comprehensive API documentation using OpenAPI/Swagger for operational visibility and future integrations.
- Successfully digitized business operations for a retail textile business, reducing manual record keeping and improving inventory visibility.
- Filed software copyright application for ArihantQInvoice Billing and Inventory Management with the Copyright Office, Government of India; application currently under scrutiny.

- **AI-Powered Portfolio & Public Profile WebApp** *Jul 2025*

HTML, CSS, JS, Bootstrap 5, GitHub Pages Project URL | GitHub

- Developed a responsive, fully client-side portfolio to showcase full-stack, AI, and Quantum projects—no frameworks or backend required.
- Integrated LinkedIn public badge sync, dark/light theme toggle, SEO OG metadata, and social profile links (GitHub, HuggingFace, Naukri, MyBharat, Credly).
- Achieved zero-cost, CI/CD-ready deployment using GitHub Pages; optimized performance and mobile-first layout.

- **RAG-Powered PDF Question Answering App** *Jul 2025*

Python, Streamlit, LangChain, HuggingFace, FAISS Live App | Hugging Face

- Engineered a document Q&A application using Retrieval-Augmented Generation (RAG), enabling users to upload PDFs and retrieve answers in plain English.
- Designed for model flexibility via HuggingFace (DeepSeek, Mixtral, Zephyr) and powered high-speed semantic chunk search using FAISS.
- Created an intuitive dark-themed Streamlit UI with support for chat and text generation modes.

- **AI Cost Optimization Advisor (Lyzr AI Architect Challenge)** *Jun 2025*

Lyzr Studio, AWS Bedrock, Claude, GPT, Gemini, Mistral Live Agent | GitHub

- Developed a GenAI agent to solve the problem of LLM cost-effectiveness by analyzing AI use cases and recommending optimal models with ROI estimation and JSON insights.
- Indexed 25+ knowledge sources with optimal chunking for RAG-based retrieval, including grounding facts to mitigate hallucination risks.
- Simulated use cases (support triage, contract summaries), achieving a documented 666% ROI in the challenge.

Technical Skills

- **Programming:** JavaScript, Python, SQL
- **Backend Development:** Node.js, Express.js, REST APIs, Microservices Architecture
- **Frontend Development:** ReactJS, Angular, HTML5, CSS3, Bootstrap, Material UI
- **Cloud Platforms:** AWS (Lambda, API Gateway, S3, Cognito, DynamoDB, CDK), GCP (Vertex AI, Gemini API)
- **Security & Authentication:** SAML, AWS Cognito, CyberArk, RSA, Secure API Design, Role-Based Access Control (RBAC)
- **Databases:** PostgreSQL, SQLite, DynamoDB
- **Data Processing & Reporting:** Python (Pandas, NumPy), Flask, Excel Automation, Data Visualization
- **DevOps & Tools:** Git, Jenkins, Postman, SoapUI, SVN, Nginx, PM2, Jira, Confluence
- **Enterprise & BFSI Exposure:** Alert Analysis Systems, Risk Monitoring Dashboards, Compliance Automation, Audit-Ready Reporting
- **AI/ML & GenAI:** LangChain, Hugging Face, FAISS, Prompt Engineering, Streamlit, OpenAI, Gemini, Claude, Mistral

Achievements

- **Completed 1920-minute** training on Microsoft Azure AI Solution (AI-102T00) by Koenig Solutions. *2024*
- **Winner Top 100** Microsoft AI Odyssey Challenge organized by Microsoft & TCS. *2024*
- **Microsoft Learn** Earned 49 Badges, 10 Trophies, 48 Modules completion and 2 Applied Skills *2024*
- **Completed multiple LinkedIn Learning** courses on Git, GitHub, React, Generative AI, CyberSecurity, Secure Coding, Data Analytics, Critical Thinking, Business Banking, Sustainability, Leadership
- **Completed national-level AI and awareness programs FutureSkills Prime (AI Applications)** – a joint initiative by **NASSCOM** & Govt. of India – and quizzes on Yoga, Cybersecurity, and Nation Building by **Ministry of Youth Affairs & Sports** and **Ministry of Defense** via MyBharat *2025*

Certifications

- **IBM** Applied Data Science with Python - Level 2
- **IBM** Big Data Foundations - Level 2
- **IBM** Artificial Intelligence Fundamentals
- **IBM** Data Fundamentals
- **Elastic** Forge The Future Hackathon
- **Elastic** Virtual Event of ElasticON Bengaluru
- **FutureSkills Prime** Applications of Artificial Intelligence
- **Microsoft** Applied Skills: Develop generative AI solutions with Azure OpenAI Service
- **Microsoft** Applied Skills: Intelligent document processing with Azure AI
- **Google Cloud** Skill Badge: GenAI Apps with Gemini & Streamlit
- **Google Cloud** Skill Badge: Explore GenAI with Vertex AI Gemini API
- **Google Cloud** Skill Badge: Gemini Multimodality and Multimodal RAG
- **Google Cloud** Skill Badge: Text Prompt Engineering Techniques
- **AWS** Cloud Practitioner Essentials (Second Edition)
- **IIT Kanpur (NPTEL)**: Introduction to Programming in C
- **IIT Kharagpur (NPTEL)**: Programming in Java
- Basics of Python (Part I & II), OOPs in Python, CS Fundamentals, and ChatGPT by **Coding Ninjas**
- Java (Basic) Certificate by **HackerRank**
- SEBI Investor Certification by **SEBI** (ID: NISM20250000203277, Valid till Jul 2027)

Extracurricular

- Actively participate in GenAI hackathons and AI webinars to stay ahead of emerging innovations. Engage with developer communities and continuously pursue short-term certifications to expand technical fluency.
- Maintain a disciplined personal routine involving endurance sports like swimming and long-distance running, building focus, mental clarity, and resilience — qualities I apply to both life and work.

Volunteering & Engagements

- **Blood Donor** Donated one unit of blood at TCS Blood Donation Camp, organized by Amrita Institute of Medical Sciences, Kochi. *Feb 2023*
- **Cycling Expedition** Participated in Environmental Awareness Bicycle Rally organized by Youth Hostels Association of India. *Dec 2015*
- **Planet Engineers 2k18** Engaged in R&D Workshop, Technothon, Quirosity, and Brainiac during LNCT's national-level tech fest. *Mar 2018*
- **TEDxLNCTBhopalWomen** Attended TEDxWomen event focused on empowerment and innovation. *Nov 2017*
- **Convolution 2018 - RGPV** Took part in Dalal Street, Quizologist, and Horcrux events at the annual university techfest. *Feb 2018*